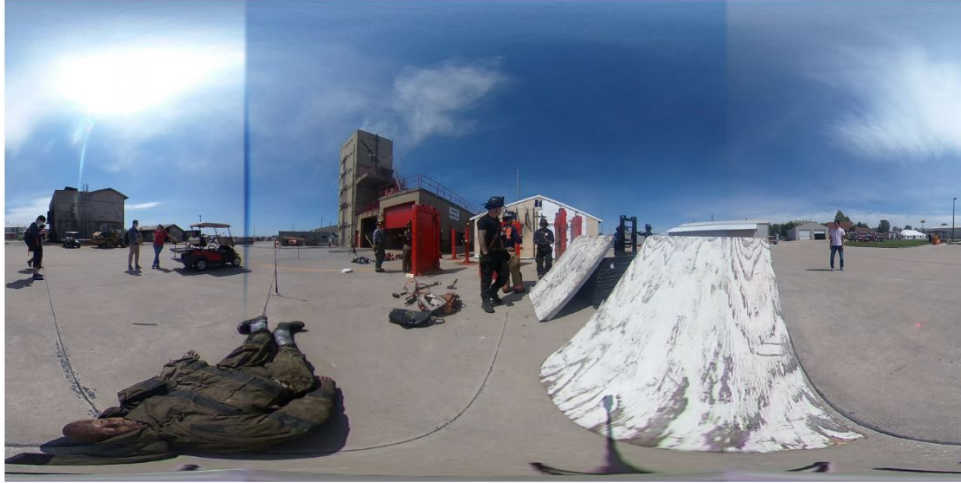


Latency-Aware 360-Degree Video Analytics Framework for First Responders Situational Awareness

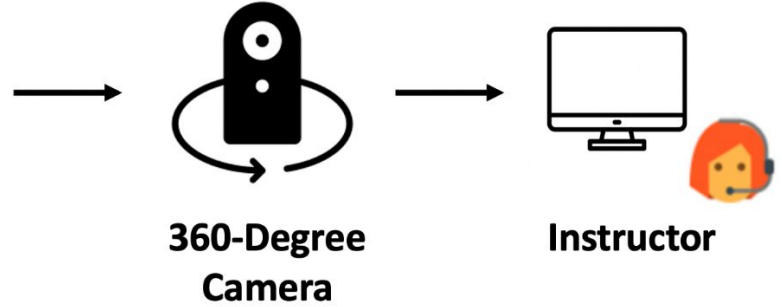
Jiaxi Li¹, Jingwei Liao², Bo Chen¹, Anh Nguyen², Aditi Tiwari¹, Qian Zhou¹,
Zhisheng Yan², Klara Nahrstedt¹



Background: First Responder Training



Training Field

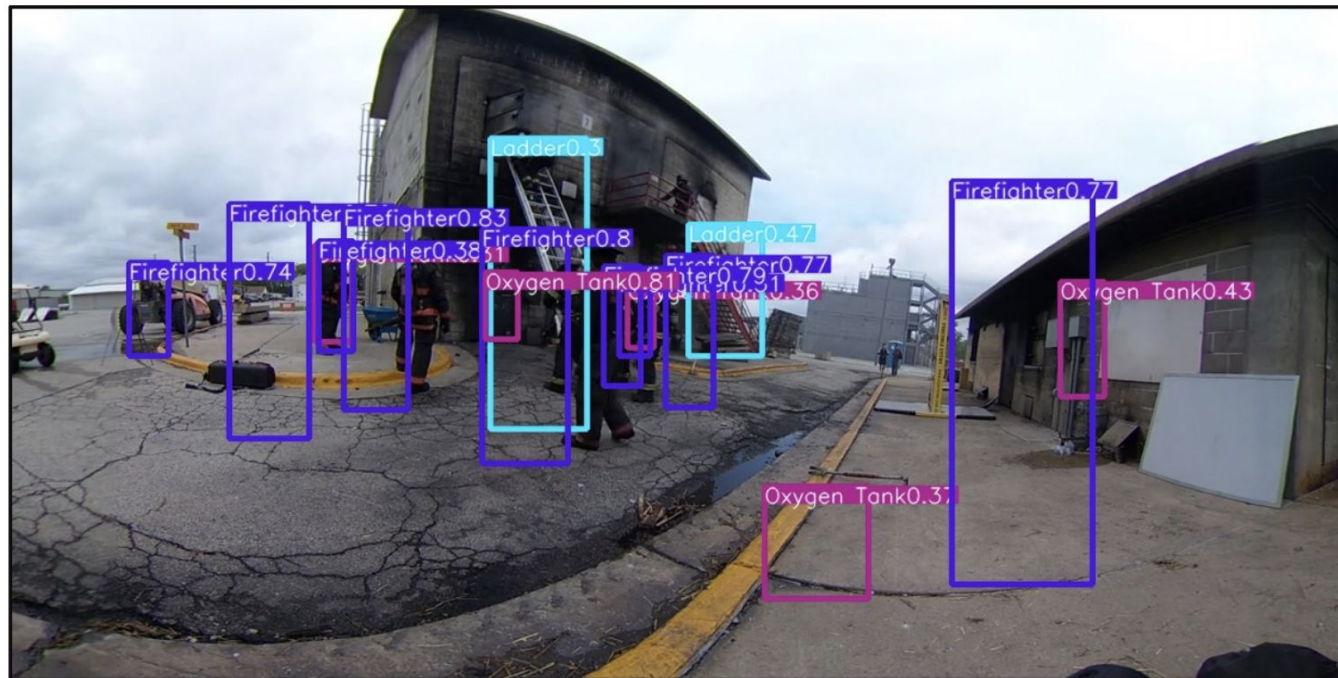


Background: First Responder Training

Object of Interest	Priority
Civilian	High
Fire	High
Smoke	High
Gas Mask	High
Firefighter	Low
Helmet	Low

Data based on an interview with a physical training instructor at Illinois Fire Service Institute (IFSI)

Viewing and Query Service



Searching for Objects

Car

Found **3 Cars** in this video:

00:33 00:50 01:23

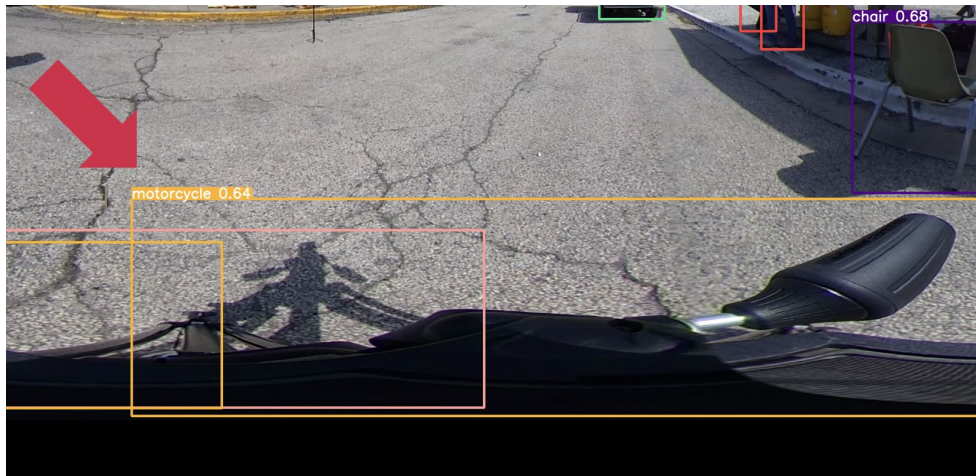
Labeling Objects

Firefighter	☒	Civilian	☐
Oxygen	☒	Ladder	☒
Helmet	☐	Car	☐

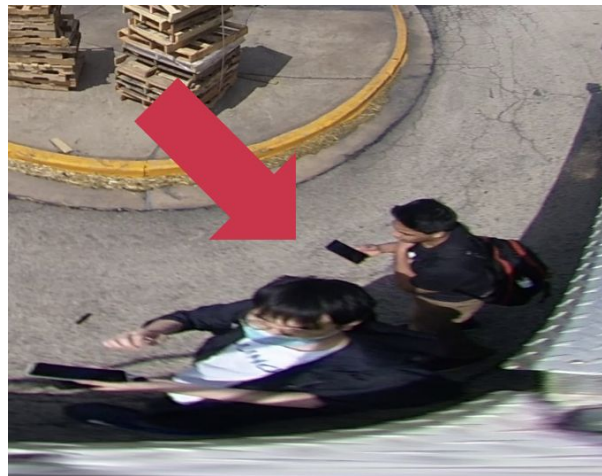
Our focus – **Object Detection**

360 Video Object Detection via 2D Object Detectors

Geometric Distortion



(Detection Error)

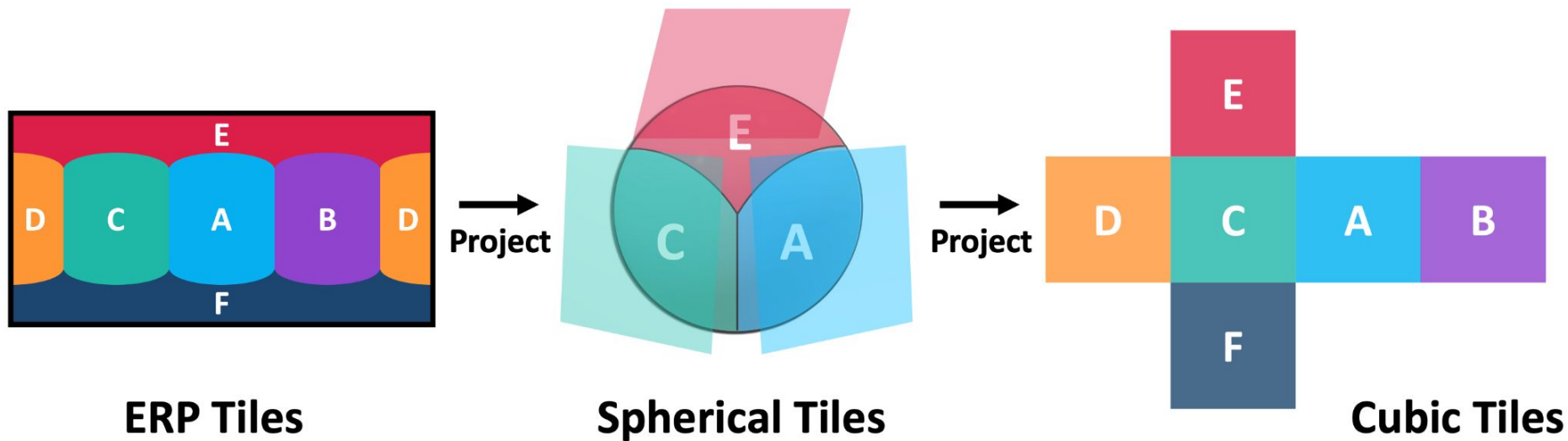


(Missed Detection)

Frame Size

- 4 times the number of pixels of a normal 1080p video

Dual-Projection Solution for 2D Object Detector Issues



Normal field of view

Dual-Projection Examples



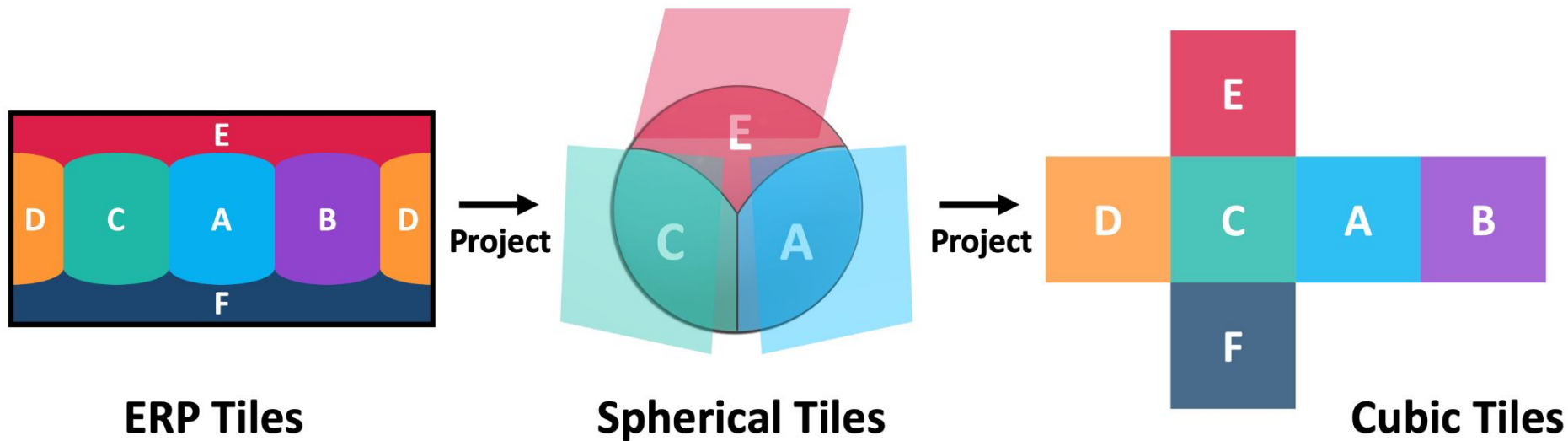
ERP Tiles

→
Dual-Projection



Cubic Tiles

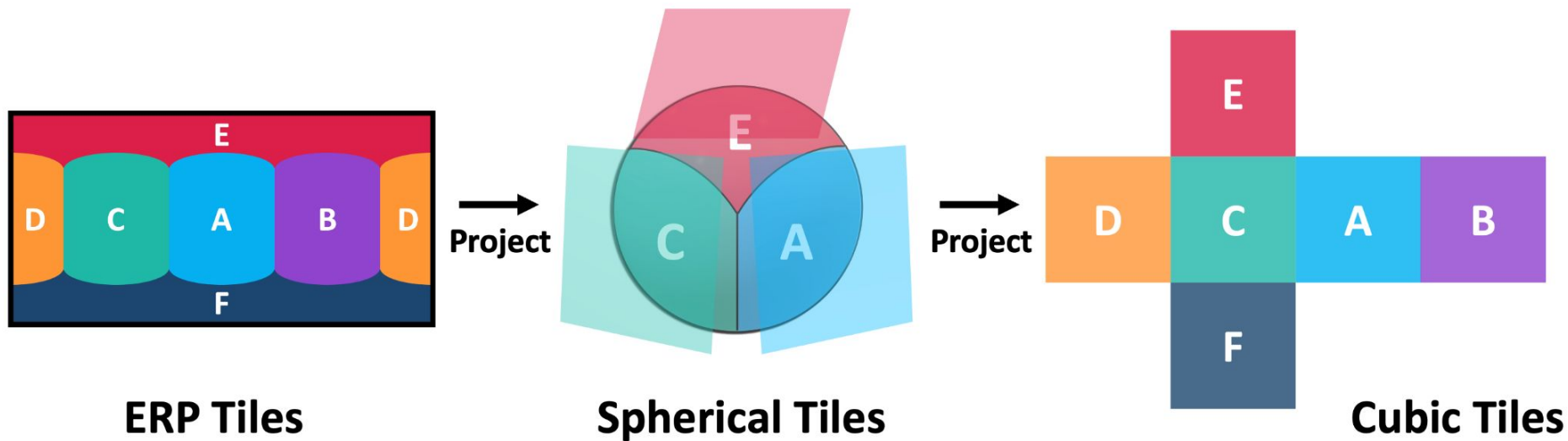
Dual-Projection Limitations



Leads to **extra processing time in transforming ERP tiles to cubic tiles**

E.g., >30% extra time on a 4-core CPU

Dual-Projection



Leads to **extra processing time in transforming ERP tiles to cubic tiles**

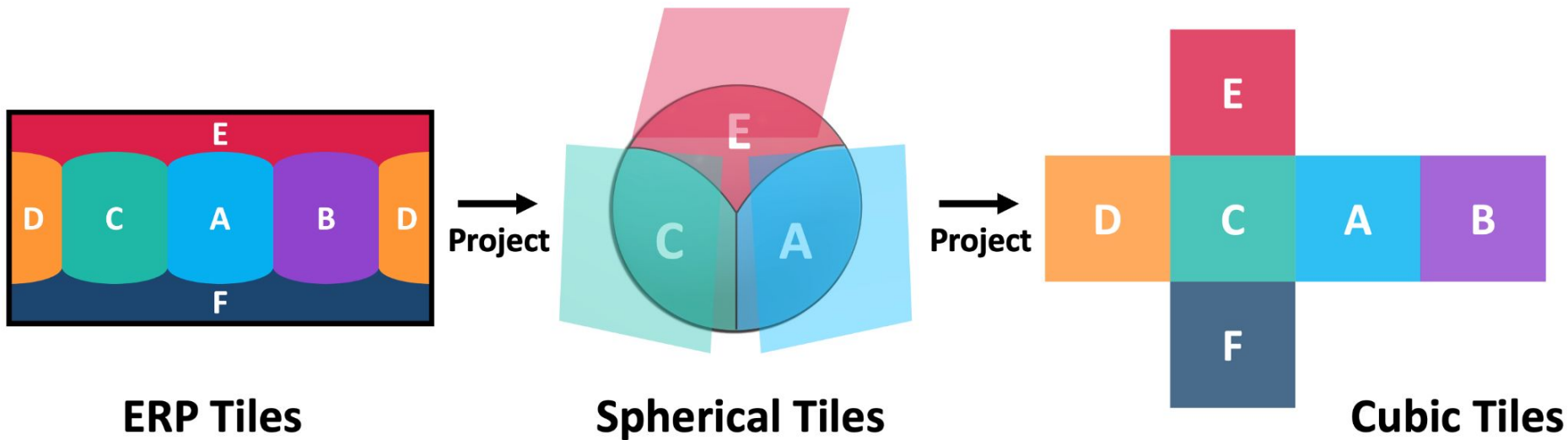
Dynamic Selection: **filters out unnecessary projection and detection.**

Dynamic Selection: Overview

Detect on **cubicTile_{i,j}** (*cubic tile j at timestamp i*) only when

- Condition (1) **cubicTile_{i,j}** is **structurally different** from **cubicTile_{i-1,j}**
- Condition (2) **cubicTile_{i-1,j}** contains a **high object cohesion**

Otherwise, inference result of the previous timestamp is adopted for **cubicTile_{i,j}**



Inter-Frame Similarity (sim): Motivation

Timestamp $i - 1$



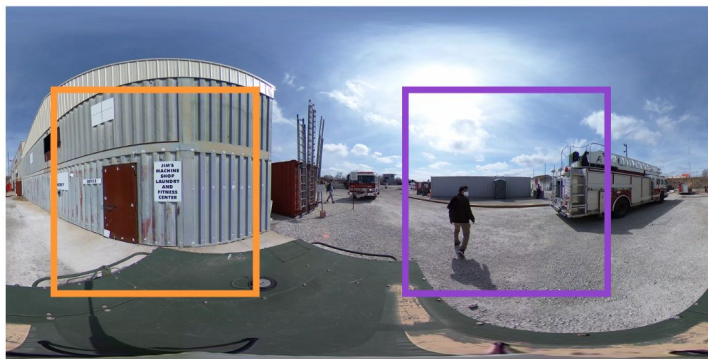
New Detection
Not Needed



New Detection
Needed



Timestamp i



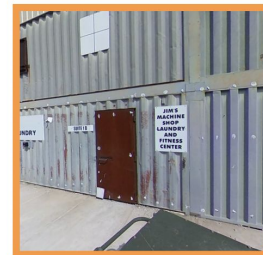
ERP tiles



cubic tiles

Inter-Frame Similarity (*sim*): Estimation

Timestamp $i - 1$

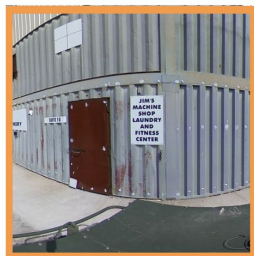


ERP tiles similarity

cubic tiles similarity

Estimate

Timestamp i



ERP tiles

cubic tiles

Inter-Frame Similarity (*sim*): Distortion Correction

$$x' = x \cdot \frac{\text{pixel sample rate}}{\text{center sample rate}} = x \cdot \frac{2\pi r \cos(\theta)}{2\pi r} = x \cdot \cos(\theta)$$



$$\text{sim}(\text{cubicTile}_{i-1j}, \text{cubicTile}_{ij}) = \text{NRMSE}(\text{erpTile}'_{i-1j}, \text{erpTile}'_{ij})$$

Object Cohesion (oc): Motivation

Timestamp $i - 1$



New Detection
Needed



New Detection
Not Needed



Timestamp i



ERP tiles



cubic tiles

Object Cohesion (oc): Formula

oc measures if any objects will remain in the *cubic tile* in the next timestamp

Factors affecting the oc:

- Number of objects
- Distance of the object to the cubic tile center (d_o)
- Object size (s_o)
- Confidence level (c_o)

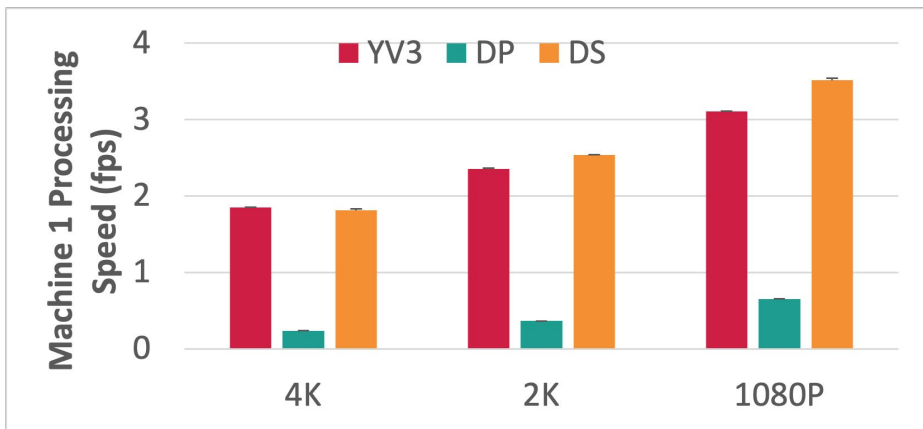
$$oc(cubicTile_{ij}) = \sum_{object\ o} \frac{c_o}{s_o \cdot d_o}$$

Models for Evaluation

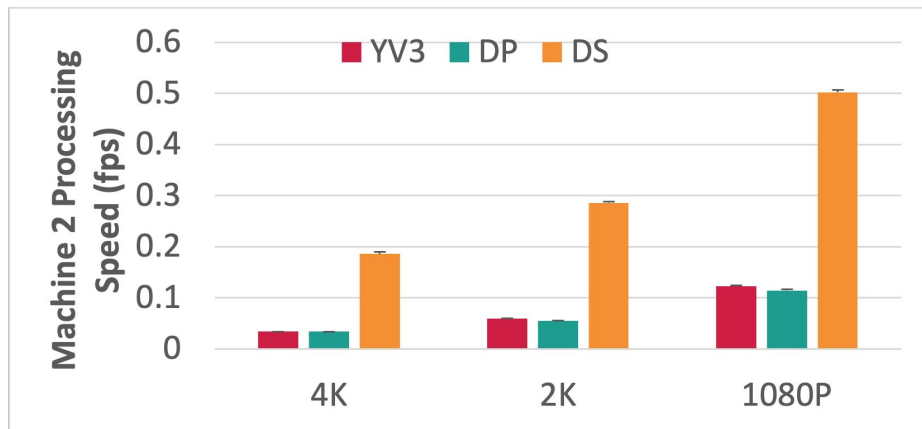
- **YV3**: Detecting using the **YOLOv3** model on **ERP** frames
- **DP**: Detecting using the **YOLOv3** model on cubic tiles generated by the **Dual-Projection** process
- **DS**: Our approach, detecting using the **YOLOv3** model on cubic tiles generated by the **Dynamic Selection** algorithm
- A dataset of 25 360-degree videos collected at Illinois Firefighter Service Institute (19 training; 6 testing)
- One 6-core CPU with a 12GB GBP and one 4-core CPU without a GPU

Evaluation of Processing Speed

NVIDIA GeForce RTX 3080 Ti



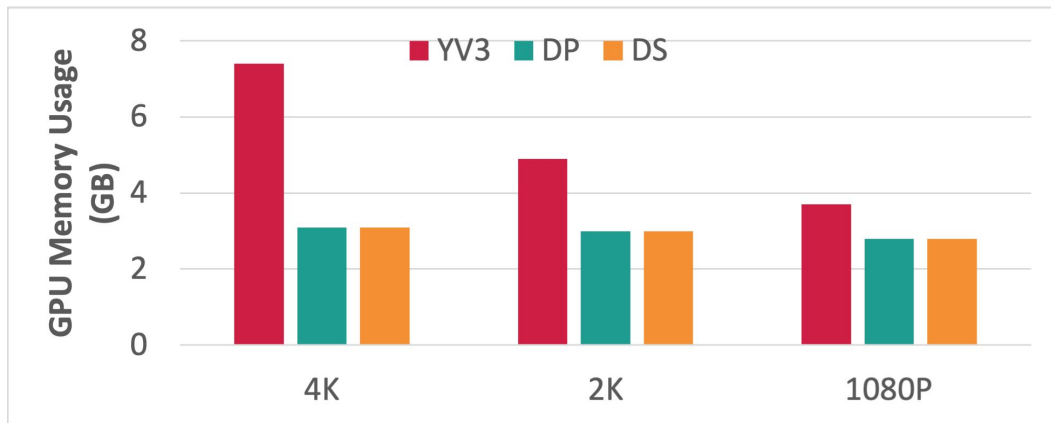
No GPU



More than 4x speed up over dual projection on GPU and non-GPU machines

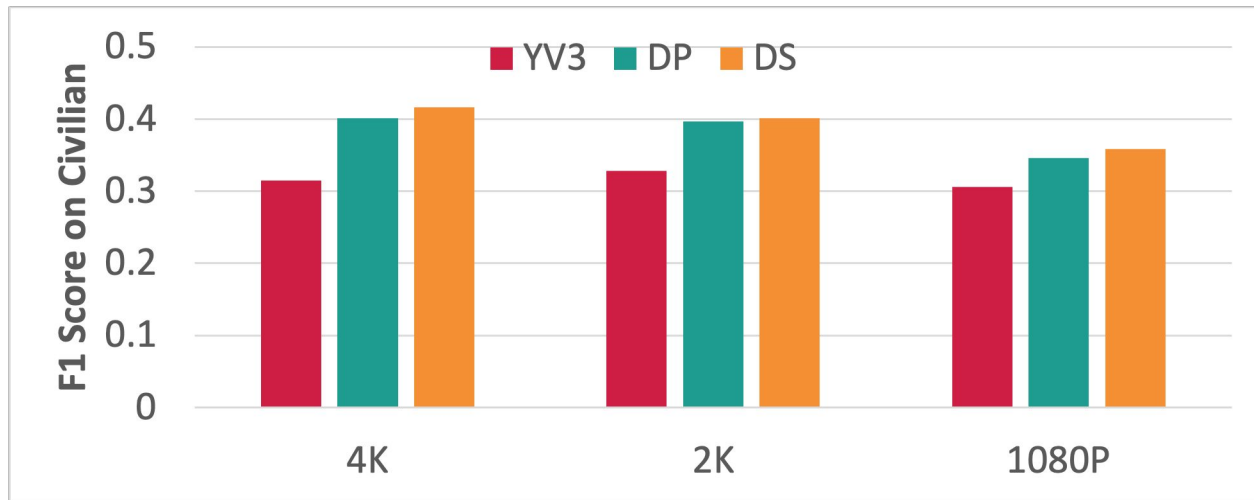
Evaluation of Memory Utilization

NVIDIA GeForce RTX 3080 Ti



Over 25% reduction in GPU memory (<4GB) compared to YV3, making high-resolution detection accessible to mid/low-end hardware

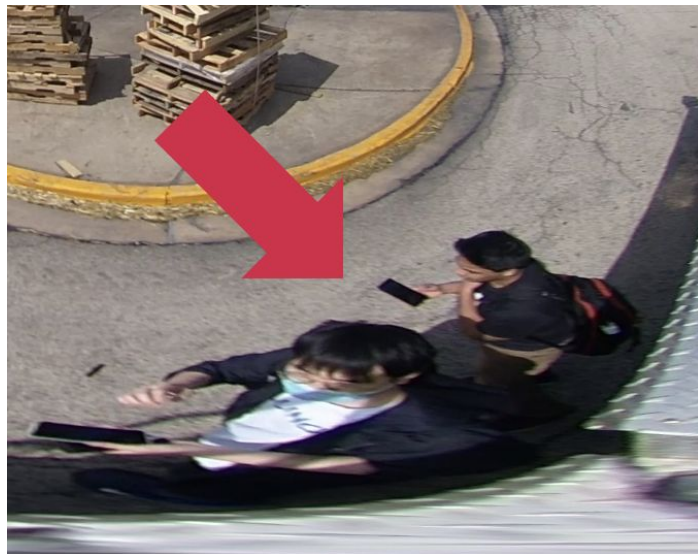
Evaluation of Detection Accuracy



Improved detection accuracy compared to YV3 (selected classes)

Conclusion

- **360 Video Object Detection Challenges**
 - Geometric Distortion
 - High Computation
- **Our Contribution: Dynamic Selection**
 - >4X Speedup
 - >25% memory usage reduction
 - Improved detection accuracy



(Missed Detection due to Distortion)

Acknowledgments

- Illinois Fire Service Institute (IFSI)
- National Science Foundation (NSF)

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